

Solution Requirements

Field	Details
Date	05-07-2026
Team ID	N/A
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Functional Requirements

Requirement ID	Requirement	Description
FR-1	User Input	User should enter Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall values •
FR-2	Data Validation	The system should validate all input values before processing •
FR-3	Crop Prediction	The ML model should predict the most suitable crop.
FR-4	Result Display	The predicted crop should be displayed clearly to the user.
FR-5	Easy Interface	The application should provide a simple and user-friendly interface.

Non-Functional Requirements

Requirement ID	Requirement	Description
NFR-1	Performance	Prediction should be generated within a few seconds.
NFR-2	Reliability	The application should provide accurate recommendations using the trained model.

Requirement ID	Requirement	Description
NFR-3	Usability	The interface should be easy to understand for farmers.
NFR-4	Scalability	Future modules like fertilizer recommendation and weather API can be integrated.
NFR-5	Maintainability	The project should be easy to update and maintain.

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB or above
- Storage: 500 MB free space
- Internet Connection (optional)

Software Requirements

- Windows 10/11
- Python 3.10+
- Flask
- Pandas
- NumPy
- Scikit-learn
- HTML
- CSS
- VS Code

Expected Outcome

The system recommends the most suitable crop based on soil nutrients and environmental conditions, helping farmers improve productivity and make informed agricultural decisions.